

FOG-ORCHESTRATOR 2.0 -- SIH EVIDENCE CURATION REPORT

Document ID: DOC-CURATION-REPORT | Date: 2026-09-28 | Audited Standard: SIH Evaluator Evidence

EVIDENCE PROVENANCE: [AUTHORITATIVE REPORT]
Complete forensic audit, file disposition rationale, and PPT claim mapping for SIH 2026-27 jury evaluation.

Smart India Hackathon (SIH 2026-27) | Problem Statement: SIH26007

Project Title: Mine-Vehicle Safety and Fleet Orchestration System for Low-Visibility and Fog Operation

Target Organization: Ministry of Steel / NMDC Limited (Bailadila Iron Ore Complex, Deposit 5)

Role: Principal Software Architect, Embedded Systems Lead & Final SIH Evidence Curator

Document Revision: 2.0 (Authoritative Freeze)

Audit Standard: Strict Evaluator-First Evidence Density & Provenance Verification

1. Executive Summary & Curation Charter

1.1 The Evaluator-First Mandate

An SIH jury evaluator examining our Google Drive link has **2 to 5 minutes** to verify the core technical claims made in our presentation. Dumping raw development histories, hundreds of debugging logs, redundant phase reports, and repetitive test outputs overwhelms the evaluator, obscures our genuine innovations, and degrades credibility.

Therefore, this curation was executed under an uncompromising standard:

1.2 Quantitative Curation Outcome

From over **530 raw files** accumulated across 10 development phases, forensic audits, and regression trials, we have distilled a **compact, high-density evidence package** of 21 authoritative documents organized into 7 clean topical folders (plus an executive root index).

Every document is provided in two synchronized formats inside [FOG-ORCHESTRATOR -- SIH EVIDENCE/](file:///c:/Users/JAGADEESH%20M/OneDrive/Documents/SIH-2026-27/FOG-ORCHESTRATOR%20%E2%80%94%20SIH%20EVIDENCE/):

- .pdf Format:** Styled for instant one-click previewing in Google Drive's native viewer.
- .md Format:** Structured Markdown for direct code repository inspection and searchability.

2. Master PPT Claim-to-Evidence Verification Matrix

This matrix allows an evaluator to immediately cross-reference every claim in the SIH presentation deck directly with its verified evidentiary proof:

3. Reconciled Calibration Truth & Numerical Integrity

Before finalizing the curated evidence package, every numerical value across all 21 documents was audited against our authoritative hardware calibration truth table to eliminate contradictions:

4. Comprehensive Audit & Disposition of Workspace Files

4.1 Retained & Curated Files (The 21 High-Value Assets)

The following 21 files form the authoritative SIH Evidence Package under FOG-ORCHESTRATOR -- SIH EVIDENCE/:

- 00_START_HERE.pdf / .md: Master evaluator navigation index.
- 01_SYSTEM_OVERVIEW/01_FOG_ORCHESTRATOR_SYSTEM_OVERVIEW.pdf / .md: Concise 2-page system overview.
- 02_ARCHITECTURE/01_SYSTEM_ARCHITECTURE.pdf / .md: Master subsystem hierarchy and boundaries.
- 02_ARCHITECTURE/02_DATA_FLOW_AND_CLOSED_LOOP.pdf / .md: Environmental, V2V, and V2I closed loops.
- 02_ARCHITECTURE/03_HMI_DATA_LINEAGE.pdf / .md: Field-by-field proof of Rule 6 (zero client physics).
- 03_HARDWARE_AND_COMMUNICATION/01_HARDWARE_ARCHITECTURE.pdf / .md: Dual ESP32 prototype chassis specs.
- 03_HARDWARE_AND_COMMUNICATION/02_VEHICLE_FIRMWARE_PARITY.pdf / .md: Side-by-side firmware comparison.
- 03_HARDWARE_AND_COMMUNICATION/03_CURRENT_WIRING_TRUTH.pdf / .md: Exact GPIO pin table and standby safety.
- 03_HARDWARE_AND_COMMUNICATION/04_V2V_V2I_COMMUNICATION_RESULTS.pdf / .md: TDM schedule and RF airtime.
- 03_HARDWARE_AND_COMMUNICATION/05_LORA_GATEWAY_RESULTS.pdf / .md: Autonomous Safe Beacon failover trials.

- 11. `04\_DIGITAL\_TWIN/01\_LIVE\_DIGITAL\_TWIN.pdf` / `.md`: Authoritative real-time operational state mirroring.
- 12. `04\_DIGITAL\_TWIN/02\_PREDICTIVE\_DIGITAL\_TWIN.pdf` / `.md`: What-if fog lookahead decision support.
- 13. `04\_DIGITAL\_TWIN/03\_DIGITAL\_TWIN\_RESULTS.pdf` / `.md`: NMDC Deposit 5 geodata benchmarks.
- 14. `05\_FOG\_SPEED\_GOVERNOR/01\_FOG\_SPEED\_GOVERNOR\_ARCHITECTURE.pdf` / `.md`: 5-constraint mathematical formulation.
- 15. `05\_FOG\_SPEED\_GOVERNOR/02\_VEHICLE\_SPEED\_CALIBRATION.pdf` / `.md`: Empirical motor calibration truth table.
- 16. `05\_FOG\_SPEED\_GOVERNOR/03\_FOG\_GOVERNOR\_RESULTS.pdf` / `.md`: Monotonic fog response policy table.
- 17. `05\_FOG\_SPEED\_GOVERNOR/04\_DRIVER\_ASSISTANCE\_LOGIC.pdf` / `.md`: In-cab cockpit warning states and driver authority.
- 18. `06\_RESULTS\_AND\_VALIDATION/01\_FINAL\_VALIDATION\_RESULTS.pdf` / `.md`: Automated test suite summary (100% pass).
- 19. `06\_RESULTS\_AND\_VALIDATION/02\_HARDWARE\_INTEGRATION\_RESULTS.pdf` / `.md`: Motor staircase and IMU tilt responses.
- 20. `06\_RESULTS\_AND\_VALIDATION/03\_FOG\_SPEED\_RESPONSE\_RESULTS.pdf` / `.md`: Closed-loop fog injection time-series.
- 21. `06\_RESULTS\_AND\_VALIDATION/04\_DIGITAL\_TWIN\_RESULTS.pdf` / `.md`: Fleet throughput retention and queue bounds.
- 22. `07\_RESEARCH\_SUPPORT/01\_SELECTED\_RESEARCH\_REFERENCES.pdf` / `.md`: 7 core statutory standards and peer-reviewed papers.

4.2 Excluded Files & Justification Matrix

Over 500 files were excluded from the evaluator Drive package to prevent cognitive overload. None of these files are lost; they remain preserved in the local development workspace:

5. Explicit Deficit & Evidence Boundaries

In adherence to strict engineering honesty (**Rule 3 and Section 23 of `AGENTS.md`**), our evidence package clearly declares what was physically verified versus what was simulated or mathematically modeled:

- 1. **Full-Scale 165.5-Tonne Haul Dumper Dynamic Execution**:
 - Status**: `MATHEMATICAL MODEL & HARDWARE-IN-THE-LOOP (HIL)`
 - Declaration**: We did not drive a real physical 165.5-tonne BEML BH100 dumper in a commercial mine. Full-scale braking distances and hydraulic retarder power curves are evaluated using validated mathematical dynamic models grounded in BEML technical specifications and SAE J1939 standards.
- 2. **Physical Microcontroller Prototype**:
 - Status**: `PHYSICAL (BENCH)`
 - Declaration**: Dual ESP32 microcontrollers, slotted optical tachometers, MPU6050 6-DOF IMUs, L298N/TB6612FNG H-bridges, and 433 MHz LoRa transceivers were physically built, instrumented, wired, calibrated, and tested on physical benches.
- 3. **Atmospheric Fog Generation**:
 - Status**: `INJECTED ENVIRONMENTAL CONDITION`
 - Declaration**: We did not build a physical pressurized aerosol chamber. Haul road visibility drops were injected via the Weather Station REST API into the Digital Twin to evaluate dynamic governor clamping.
- 4. **Autonomous Emergency Braking**:
 - Status**: `DELIBERATELY REJECTED AS AN ARCHITECTURAL RISK`
 - Declaration**: The system does NOT take autonomous control of braking or steering. It provides in-cab driver guidance, visual/audible alarms, and advisory dispatch pacing. The licensed human operator remains in full physical authority.

6. Hostile Evaluator Verification (The 5-Minute Test)

Pretending to be a skeptical SIH evaluator with only 5 minutes, here is how each critical question is answered in seconds:

- 1. **What did the team actually build?**
Open `01\_SYSTEM\_OVERVIEW/01\_FOG\_ORCHESTRATOR\_SYSTEM\_OVERVIEW.pdf`. Complete summary of physical prototype, 3D Digital Twin, and driver-assistance HMI.
- 2. **Is the hardware real or just a simulation?**
Open `03\_HARDWARE\_AND\_COMMUNICATION/01\_HARDWARE\_ARCHITECTURE.pdf` and `03\_CURRENT\_WIRING\_TRUTH.pdf`. Shows physical ESP32 chassis, optical encoders, GPIO pinouts, and TB6612FNG standby wiring.
- 3. **Does the fog-to-safe-speed governor actually work?**
Open `05\_FOG\_SPEED\_GOVERNOR/03\_FOG\_GOVERNOR\_RESULTS.pdf`. Complete table showing safe speed dropping monotonically from 0.80 m/s down to 0.08 m/s as fog intensifies.

4. ***Does the vehicle autonomously slam on the brakes?***

\$\to\$ Open `05\_FOG\_SPEED\_GOVERNOR/04\_DRIVER\_ASSISTANCE\_LOGIC.pdf`. Proves the human operator remains in authoritative command; system provides cockpit guidance and alerts.

5. ***What happens if Wi-Fi disconnects in the pit?***

\$\to\$ Open `03\_HARDWARE\_AND\_COMMUNICATION/05\_LORA\_GATEWAY\_RESULTS.pdf`. Proves autonomous failover to 1.0 Hz 433 MHz LoRa Safe Beacon within 24.5 ms of link loss.

6. ***What measurable productivity results did the team achieve?***

\$\to\$ Open `06\_RESULTS\_AND\_VALIDATION/04\_DIGITAL\_TWIN\_RESULTS.pdf`. Shows 68.4% throughput retention under fog vs 0% (total shutdown) in conventional baseline operation.

7. Final Clean Evidence Room Directory Structure

FOG-ORCHESTRATOR -- SIH EVIDENCE/

?-- 00_START_HERE.pdf / .md	# Master Evaluator Navigation & Claim Matrix
?-- 01_SYSTEM_OVERVIEW/	
+-- 01_FOG_ORCHESTRATOR_SYSTEM_OVERVIEW.pdf / .md	# Concise 2-Page Executive Overview
?-- 02_ARCHITECTURE/	
?-- 01_SYSTEM_ARCHITECTURE.pdf / .md	# Subsystem Hierarchy & Block Architecture
?-- 02_DATA_FLOW_AND_CLOSED_LOOP.pdf / .md	# Environmental, V2V & V2I Closed Loops
+-- 03_HMI_DATA_LINEAGE.pdf / .md	# Proof of Zero Client Physics (Rule 6)
?-- 03_HARDWARE_AND_COMMUNICATION/	
?-- 01_HARDWARE_ARCHITECTURE.pdf / .md	# Dual ESP32 Prototype Specifications
?-- 02_VEHICLE_FIRMWARE_PARITY.pdf / .md	# Vehicle A vs Vehicle B Firmware Parity
?-- 03_CURRENT_WIRING_TRUTH.pdf / .md	# Complete GPIO Pinouts & Standby Safety
?-- 04_V2V_V2I_COMMUNICATION_RESULTS.pdf / .md	# TDM Schedule, 11-Field V2V & Airtime
+-- 05_LORA_GATEWAY_RESULTS.pdf / .md	# Emergency Safe Beacon Failover Trials
?-- 04_DIGITAL_TWIN/	
?-- 01_LIVE_DIGITAL_TWIN.pdf / .md	# Real-Time Operational State Mirroring
?-- 02_PREDICTIVE_DIGITAL_TWIN.pdf / .md	# What-If Fog Decision Support
+-- 03_DIGITAL_TWIN_RESULTS.pdf / .md	# NMDC Deposit 5 Geodata & Determinism
?-- 05_FOG_SPEED_GOVERNOR/	
?-- 01_FOG_SPEED_GOVERNOR_ARCHITECTURE.pdf / .md	# 5-Constraint Physics Formulation
?-- 02_VEHICLE_SPEED_CALIBRATION.pdf / .md	# Optical Tachometer Truth Table (K_cal = 34.58)
?-- 03_FOG_GOVERNOR_RESULTS.pdf / .md	# Monotonic Safe Speed Policy Table
+-- 04_DRIVER_ASSISTANCE_LOGIC.pdf / .md	# In-Cab Cockpit Guidance & Driver Authority
?-- 06_RESULTS_AND_VALIDATION/	
?-- 01_FINAL_VALIDATION_RESULTS.pdf / .md	# Automated Test Summary (100% Pass)
?-- 02_HARDWARE_INTEGRATION_RESULTS.pdf / .md	# Motor Staircase & IMU Angle Responses
?-- 03_FOG_SPEED_RESPONSE_RESULTS.pdf / .md	# Closed-Loop Telemetry Response Traces

| +-- 04_DIGITAL_TWIN_RESULTS.pdf / .md # Fleet Throughput & Collision Prevention Proof

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+-- 07_RESEARCH_SUPPORT/

+-- 01_SELECTED_RESEARCH_REFERENCES.pdf / .md # 7 Grounding Industry Standards & Papers

Certified by Team SYNQRA for SIH 2026-27 Jury Evaluation. All evidence verified against active codebase and physical test benches.